# PPT SCRIPT

## ACT 1 — Setup *(0:00 — 1:35)*

### Slide 1 — Portada *(0:00 — 0:10)*

> Good morning. My name is Francisco Sandoval, and the project I'm presenting today is *Word-Level ASL Recognition from Hand Landmarks*, supervised by Dr. Moisés Martínez.

[**0:10**]

### Slide 2 — Agenda *(0:10 — 0:25)*

> The presentation is organised in eight blocks — from the problem and the goals, through the architecture, the technical work, and the results, ending with the ethical analysis and the conclusions. Let me start with the problem.

[**0:25**]

### Slide 3 — Introduction · Problem *(0:25 — 1:00)*

> Sign languages are natural languages with their own visual-spatial structure — not simple gesture codes. Their meaning emerges from hand shape, motion, orientation, location, and timing.
>
> This makes automatic recognition from video a demanding problem. **The same sign varies across signers, viewpoints, execution speeds, and recording conditions** — and that variability limits the reach of every accessibility tool that depends on it.
>
> This project sits exactly at that intersection: accessibility, computer vision, and the modelling of how something changes through time.

[**1:00**]

### Slide 4 — Main Goal + Hypothesis (texto) *(1:00 — 1:25)*

> The main goal is to design, implement, and experimentally validate a research prototype for word-level ASL recognition from short video clips, using **hand-landmark representations** — which I'll explain in a moment — and temporal neural models.
>
> The research question that drives the entire thesis is whether hand-landmark sequences provide enough information for a temporal model to classify short ASL video clips into word glosses.

[**1:25**]

### Slide 4 build-up — Pipeline visual *(1:25 — 1:50)*

> The system works like this. *(click — appears video clip)* A short video of a sign is captured. *(click — appears landmarks on hand)* From every frame, we extract twenty-one three-dimensional hand keypoints using **MediaPipe Hands**, a public hand-tracking library from Google — so instead of feeding raw pixels into the model, we feed the geometry of the hand. *(click — appears Top-K)* That sequence is then classified into a gloss — the dictionary form of an ASL word — with top-K confidence.
>
> **The whole project explores whether geometry is enough.**

[**1:50**]

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## 🟩 ACT 2 — Project framing *(1:50 — 3:30)*

### Slide 5 — Specific Goals (con build-up 6 clicks) *(1:50 — 2:25)*

> The main goal decomposes into six specific objectives.
>
> *(SO1 in)* — analyse the technical and scientific background of sign language recognition. *(SO2 in)* — define the landmark-based data representation. *(SO3 in)* — compare candidate temporal architectures under matched conditions. *(SO4 in)* — validate the champion model with statistically meaningful protocols. *(SO5 in)* — verify generalisation through cross-dataset evaluation and vocabulary scaling. *(SO6 in)* — build a runtime prototype and evaluate the ethical implications.
>
> Each of these is validated experimentally in the next blocks.

[**2:25** · *si vas con prisa, hazlos en 2 clicks de 3 cada uno*]

### Slide 6 — Validation Methods *(2:25 — 2:50)*

> The validation strategy has three tracks. **Quantitative validation** with multi-seed runs, stratified 5-Fold cross-validation, and standard metrics — Top-1, Top-5, Macro-F1, and Balanced Accuracy. **Cross-dataset validation**, retraining the exact same pipeline on a second ASL benchmark with no architectural changes. And a **qualitative validation protocol** designed for the runtime prototype, which is in progress.

[**2:50**]

### Slide 7 — Work Plan (7 clicks por fase) *(2:50 — 3:30)*

> The work plan ran across five months in 2026, organised in seven phases.
>
> *(Phase A — Project Framing)* Phase A was the proposal — scope definition and the initial pipeline approved. *(Phase B — Hand Detection · pivot)* Phase B explored static letter recognition with a YOLO hand detector. After two weeks, the bottleneck was clear: **static images cannot capture the temporal nature of signing.** *(Phase C — Landmark*

> *Baseline)* That triggered the methodological pivot, and Phase C established the WLASL30 baseline with MediaPipe and the GRU+Attention model that became the champion. *(Phase D — Vocabulary Scaling & Ethics)* Phase D extended the work with vocabulary scaling and the ethical framework. *(Phase E — Mobile Prototype)* Phase E built the mobile prototype — Expo client and FastAPI server. *(Phase F — Modular Refactor)* Phase F refactored the experimental code into a production package and added MS-ASL50. *(Phase G — Documentation)* And Phase G is documentation.

[**3:30**]

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## 🟨 ACT 3 — System design *(3:30 — 4:30)*

### Slide 8.1 — Training Architecture *(3:30 — 4:00)*

> The system has two architectures — **a training architecture, offline, that produces a checkpoint**, and an inference architecture, online, that serves predictions in real time.
>
> The training side starts with raw video datasets — WLASL and MS-ASL — flows through MediaPipe landmark extraction, into the GRU+Attention model trained under a multi-seed protocol, and produces a checkpoint, `best_model.pt`. Everything is tracked: YAML configs, git commit, metrics, and confusion matrices live in experiment manifests, which is what makes every run reproducible.

[**4:00**]

### Slide 8.2 — Inference Architecture *(4:00 — 4:30)*

> The inference side is the same model in production. The mobile client captures a short clip, sends it to the FastAPI server, which extracts landmarks frame by frame and runs the GRU+Attention model. The server returns the top-K prediction — for example, "no" with eighty-seven percent confidence, "before" with six, "cool" with three. Top-K matters because it lets the user make the final call instead of accepting a single forced prediction.

[**4:30**]

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## 🟧 ACT 4 — Technical Development *(4:30 — 4:50)*

### Slide 9 — Technical Development (mosaico 2x3) *(4:30 — 4:50)*

> The experimental work is structured in six blocks: **dataset preparation** with MediaPipe and twenty-one keypoints per frame; **architecture selection** on WLASL30; a **sequence-length sweep** to pick the input length; **multi-seed validation** of the champion; **cross-**

> **dataset validation** on MS-ASL50; and **vocabulary scaling** on WLASL from thirty to fifteen hundred classes.
>
> Let's look at the results.

[**4:50**]

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## 🟥 ACT 5 — Results · the centre of gravity *(4:50 — 7:40)*

### Slide 10.1 — Research Question sola *(4:50 — 5:05)*

> The research question we set out to answer was this — **can hand-landmark sequences provide enough information for a temporal model to classify short ASL video clips into word glosses?**

[**5:05**]

### Slide 10.2 — Facts (4 experimentos) *(5:05 — 5:20)*

> To answer it, we ran four blocks of experiments. Architecture comparison on WLASL30. Statistical validation under matched conditions. Cross-dataset generalisation on MS-ASL50. And vocabulary scaling on WLASL — from thirty to fifteen hundred classes.

[**5:20**]

### Slide 10.3 — Headline Numbers *(5:20 — 5:40)*

> Before going into detail, the three headline numbers. The WLASL30 champion reaches **63.85 percent Top-1** accuracy under 5-Fold cross-validation. The same pipeline trained on MS-ASL50, with no architectural changes, reaches **70.20 percent**. And the landmark detection rate on MS-ASL50 is eighty-four percent — high enough to support clean training.

[**5:40**]

### Slide 10.4 — Fact 1 · Architecture Selection *(5:40 — 6:05)*

> First, architecture selection. Five candidates trained under matched conditions on WLASL30 — LSTM, plain GRU, GRU with hand-engineered features, GRU with Temporal Attention, and a Transformer.
>
> The GRU with **Temporal Attention** dominates — fifty-nine Top-1, ninety-one Top-5. The attention mechanism is what lets the model focus on the few key frames where the linguistic content concentrates, ignoring the neutral preparation and recovery frames around them. The Transformer underperforms here because at this data scale, recurrent

> inductive biases pay off — the Transformer's advantage only appears at much larger corpus sizes.

[**6:05**]

## Slide 10.5 — Fact 2 · Statistical Validation *(6:05 — 6:30)*

> Second, statistical validation under two independent protocols. The 5-Fold cross-validation — averaging across non-overlapping test partitions — gives **63.85 percent plus-minus 4.5 percent Top-1**. This is the headline number of the thesis, and the one most comparable to the sign language recognition literature.
>
> A second protocol — multi-seed runs on a fixed split — confirms training stability. And the sequence-length sweep selected sixty-four frames as the input length, **not because it maximises any single metric, but because it produces the most reproducible training across seeds.** The selection criterion is reproducibility, not peak accuracy.

[**6:30**]

## Slide 10.6 — Fact 3 · Cross-Dataset *(6:30 — 7:00)*

> Third — and this is the most important slide of the presentation — cross-dataset generalisation. The same code path, the same hyperparameters, just pointed at a second ASL benchmark: MS-ASL50, fifty classes, almost two thousand clips.
>
> The result is **70.20 percent Top-1**. But the number that matters here is not the accuracy — it is the **standard deviation**. **2.94 versus 2.89 percentage points**, across test sets that differ by sevenfold in size.
>
> **Training stability is a property of the pipeline, not of a specific benchmark.** The pipeline does not need dataset-specific tuning to produce reproducible results on a second ASL corpus.

[**7:00**]

## Slide 10.7 — Fact 4 · Vocabulary Scaling (paralelas) *(7:00 — 7:25)*

> Fourth — vocabulary scaling. We retrained the champion on progressively larger WLASL subsets, from thirty classes up to fifteen hundred. This is a parallel-coordinates view — each line is one experiment, coloured from green at thirty classes to red at fifteen hundred.
>
> Look at the signature. Train accuracy on the fourth axis **stays high across the entire range**, even at fifteen hundred classes the model reaches seventy-five percent on training. But every validation and test axis collapses. **The train side stays flat while every test-side axis crashes — this is the textbook fingerprint of overfitting driven by sample scarcity, not by architectural failure.** At fifteen hundred classes there are only four-and-a-half videos per class. The bottleneck is data, not the model.

[**7:25**]

### Slide 10.8 — Research Question · Answer *(7:25 — 7:40)*

> So coming back to the research question — **yes**, within the tested vocabulary scale, hand-landmark sequences support reproducible word-level ASL recognition, with stability that holds across two independent benchmarks.

[**7:40**]

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## 🟪 ACT 6 — Ethical Analysis *(7:40 — 8:20)*

### Slide 13.1 — Scores by Principle *(7:40 — 8:00)*

> The ethical analysis uses Google's AI Principles framework — twenty-six metrics scored across seven principles. The strongest pillars are **scientific excellence** at ten out of ten and **safety** at almost nine. The weakest is **privacy**, at five out of ten, driven entirely by the inference pipeline — where the server still processes raw video frames. The overall score is **seven-point-six out of ten**.

[**8:00**]

### Slide 13.2 — Key Insight *(8:00 — 8:20)*

> But the key insight of the analysis is that **one architectural decision propagates positive effects across multiple ethical principles**. Choosing MediaPipe landmarks as input — originally a technical choice driven by data scarcity — reduces appearance-based bias because the model never sees skin tone or background, implements data minimisation by construction because sixty-three numbers per frame replace a full image, and limits misuse potential because abstract landmarks cannot be repurposed for surveillance. **A technical decision became three ethical wins, almost for free.**

[**8:20**]

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## 🟦 ACT 7 — Closing *(8:20 — 9:15)*

### Slide 14.1 — General Conclusion *(8:20 — 8:40)*

> Coming back to the general goal of the project — the central conclusion is that hand-landmark sequences combined with a GRU+Attention temporal model can support feasible word-level ASL recognition. The qualifier that matters is **reproducible**. The pipeline holds across two independent ASL benchmarks with virtually identical standard deviations, despite a sevenfold difference in test-set size.

[**8:40**]

## Slide 14.2 — Specific Objectives (recap) *(8:40 — 9:00)*

> Going through the six specific objectives — the technical background covered in the State of the Art; the landmark representation defined and validated; five architectures compared with GRU+Attention selected; statistical validation under two independent protocols; cross-dataset generalisation to MS-ASL50 and scaling to fifteen hundred classes; and the runtime prototype on mobile, together with the ethical analysis. **All six objectives delivered with experimental evidence.**

[**9:00**]

## Slide 14.3 — Next Steps *(9:00 — 9:15)*

> Three natural extensions. **Qualitative validation** with native ASL signers, using the protocol already designed. **Self-supervised landmark pretraining** on the full WLASL2000 corpus, to break the per-class data limit identified in the scaling experiment. And **multi-modal extensions**, incorporating facial expression and body pose to capture the non-manual markers that signing relies on and that the current pipeline ignores by design.

[**9:15**]

## Slide 15 — Thank You *(9:15 — 9:20)*

> Thank you. I'm happy to take your questions.

[**9:20** · *45 segundos de margen para improvisar, pausas, o transiciones lentas*]